Name: ______________________________ Tutorial group: ________________

Matrикulation number: | | | | | | | | | |

30 October 2021 **MH1812 Test 2** 60 minutes

QUESTION 1. **(20 marks)**

Solve the linear recurrence relation $a_n = a_{n-1} + 6a_{n-2}$ with initial values $a_0 = 5, a_1 = 0$.

Solution.

The characteristic equation is:

$$x^2 - x - 6 = 0$$
$$(x-3)(x+2) = 0$$

then $x = -2$ and $x = 3$ are two distinct roots of the equation.

We have: $a_n = u3^n + v(-2)^n$ then

$$5 = a_0 = u + v$$
$$0 = a_1 = 3u - 2v$$

hence $u = 2, v = 3$.

Therefore, $a_n = 2.3^n + 3.(-2)^n$.

QUESTION 2. **(20 marks)**

We are given 4 rooks, 2 of which are red and 2 of which are blue. In how many ways can the 4 rooks be placed on a 6-by-6 board so that no two rooks can attack each together? (Two rooks attack each other if they are in the same row or the same column.)

Your answer should be an explicit integer and must not be an unevaluated or partially evaluated expression.

Solution.

If all the rooks are distinguishable:

the 1st rook has 6^2 choices for any places on the board,

for each choice of the 1st rook, the 2nd rook has 5^2 choices after removing the row and column of 1st rook,

for each choice of 1st and 2nd rook, the 3rd rook has 4^2 choices,

for each choice of 1st, 2nd, and 3rd rook, the 4th rook has 3^2 choices.

Since 2 rooks are red and 2 rooks are blue, we need to remove 2!2! indistinguishable patterns, then the number of ways is $\dfrac{6^2.5^2.4^2.3^2}{2!2!} = 32400$.

Special question: Only rooks of same color and in the same row or same column can attack each other. First, we choose the places for red rooks by choosing 2 columns and 2 rows crossing each others and place 2 reds in the 2 diagonal intersections. There are $C_2^6.C_2^6.2 = 450$ possible ways. For each case of placing 2 reds, the 2 blue rooks can be places following 3 cases:

Case 1. 2 blues placed on different rows than red.

We choose 2 out of 4 remaining rows and 2 out of 6 columns to form 4 intersections, and place the blues in the 2 diagonal points. There are $C_2^4.C_2^6.2 = 180$ ways.

Case 2. Blue and red share one common row.

We choose 1 out of 2 red rook rows, choose 1 out of 4 without red rook rows, choose 1 out of 5 columns (except the column already taken by 1 red), and 1 out of 5 columns (except the column just chosen). There are $C_1^2.C_1^4.C_1^5.C_1^5 = 200$ ways.

Case 3. Blue and red share 2 common rows. If the first blue is aligned with the second red, the second blue has 5 choices, and if it is not aligned with the second red, the first blue has 4 choices, and the second blue has 4 choices. There are $5 + 4.4 = 21$ ways.

Total possible ways is $450.(180 + 200 + 21) = 180450$.

For graders only	Question	1	2	3(a)	3(b)	3(c)	4(a)	4(c)	Bonus	Total
	Marks									

QUESTION 3. **(30 marks)**

Give the set $A = \{1, 2, 3, 4\}$, and a relation R defined on A as for all $x, y \in A$, $(x, y) \in R$ if and only if $|x - y| = 2$.

(a) (10 points) Express R explicitly by listing out all its elements.

$$R = \{(1,3), (3,1), (2,4), (4,2)\}$$

(b) (12 points) Is R reflexive? Symmetric? Anti-symmetric? Transitive? (Justify your answer)

1. R is reflexive since $(1,1)$ is not in R
2. R is symmetric since $(1,3)$ and $(3,1)$ are in R, $(2,4)$ and $(4,2)$ are in R.
3. R is not anti-symmetric since $(1,3)$ and $(3,1)$ are in R but $1 \neq 3$.
4. R is not transitive since $(1,3)$ and $(3,1)$ are in R but $(1,1) \notin R$.

(c) (8 points) Find R^t.

$$R^t = \{(1,3), (3,1), (1,1), (3,3), (2,4), (4,2), (2,2), (4,4)\}$$

QUESTION 4. **(30 marks)**

(a) (20 points) Prove using membership table that for any sets A, B, C

$$(A \cup B) - C = (A - C) \cup (B - C)$$

Solution

A	B	C	$A \cup B$	$A - C$	$B - C$	$(A \cup B) - C$	$(A - C) \cup (B - C)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	0	1	1	1
0	1	1	1	0	0	0	0
1	0	0	1	1	0	1	1
1	0	1	1	0	0	0	0
1	1	0	1	1	1	1	1
1	1	1	1	0	0	0	0

Since all the entries of $(A \cup B) - C$ and $(A - C) \cup (B - C)$ are the same when A, B, C are the same, $(A \cup B) - C = (A - C) \cup (B - C)$.

(b) (10 points) Given two sets S, T with $S \subseteq T$ and $|S| = 1812, |T| = 2021$, how many X are there such that $S \subseteq X \subseteq T$?

Solution There are 209 elements in T but not in X, each element has 2 choices, in X or not in X, then there are 2^{209} such X.

BONUS QUESTION. **(10 marks)**

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

For how many integers from 1 through 99999 is the sum of their digits equal to 13 ?

Solution

Let x_1, x_2, x_3, x_4, x_5 be the digit of such integers. We have

$$x_1 + x_2 + x_3 + x_4 + x_5 = 13, x_1, x_2, x_3, x_4, x_5 \geq 0.$$

Hence $(x_1+1)+(x_2+1)+(x_3+1)+(x_4+1)+(x_5+1) = 17$, where $x_1+1, x_2+1, x_3+1, x_4+1, x_5+1 \geq 1$8(1)

The number of solution of (1) is equivalent to the number of strings of 18 number one and 4 number 0 of form 1...101..101..101..101..1, where the number of 1 between two 0s is the corresponding $x_i + 1$. Since there are 17 possible positions to place 0, there are C_4^{17} solutions of equation (1). However, we need to exclude the cases

$(10, 0, 0, 0, 3)$ and its permutation
$(10, 0, 0, 2, 1)$ and its permutation
$(10, 0, 1, 1, 1)$ and its permutation
$(11, 0, 0, 0, 2)$ and its permutation
$(11, 0, 0, 1, 1)$ and its permutation
$(12, 0, 0, 0, 1)$ and its permutation
$(13, 0, 0, 0, 0)$ and its permutation

Hence, the possible numbers is $C_4^{17} - \frac{5!}{3!} - \frac{5!}{2!} - \frac{5!}{3!} - \frac{5!}{3!} - \frac{5!}{2!2!} - \frac{5!}{3!} - \frac{5!}{4!} = 2205.$